

Anonymous DHT Profiling and the Equalization Contract

Turning transcript skews into actionable odds – inflation and identification budgets

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Abstract

Anonymous DHT lookups are routinely “anonymized” by delegation (random walks), split queries, cover traffic, and padding. Yet practical deployments still enable *profiling*: a passive observer (or a small set of relays) aggregates many lookup transcripts and infers which keys, content IDs, or destinations a user is querying.

This note makes profiling *budgetable*. We introduce an *equalization contract*: a deployment promises that a chosen transcript projection (contacts, timing buckets (bucket design: Synthesis 27)[8], retry patterns, congestion witnesses, ...) is nearly key-independent. Using the series’ odds – inflation bridge, we show how such skews compose over repeated lookups into explicit multiplicative posterior bounds and into identification sizing rules. The point is operational: an implementer can (i) decide *what* to equalize, (ii) certify the contract on logs, and (iii) publish a compact “privacy label” that downstream consumers can reason about.

Scope. We formalize a deployable *equalization contract* for a chosen transcript projection $X = \pi(T)$: across keys/classes $k \in \mathcal{K}$, the law of X is nearly key-independent in a max-divergence sense. The contract is designed to feed directly into odds-inflation / MaxL endpoints and to compose over a declared repetition window.

Units. Budgets are measured in bits: we use $\log_2$ and 2^ϵ for odds-inflation / pointwise-maximal-leakage style caps (see Synthesis 8).[32]

Imports. Odds-inflation endpoints and sizing rules are imported from Anonymity A and the endpoint-bridge synthesis note.[2, 3] Projection naming (observation tiers, exposure normal forms, ENF-ID) is imported from Synthesis 3, and repetition semantics (TW declaration and TW-ID) from Synthesis 4.[11, 12] Retries and pinning rules are imported from the schedule-only wrapper (Operational A), and state-dependent accounting from State 1.[4, 5] For stop-time padding and deadline normal forms, see the Stop-Time Padding Addendum.[18] For log-audit menu-level certification (many keys/classes/projections) see Synthesis 25, and for repeated/expanding audits (change-control-safe statistical confidence over time) see Synthesis 26.[6, 9] For large-alphabet projections (feature/hash spaces) and receipt-facing coarsening rules, see Synthesis 28; for data-dependent selection of maps (bucket boundaries, top- m sets), see Synthesis 29.[7, 39]

Exports. Exports an equalization contract for a declared X : (i) a pairwise (η, δ) max-divergence budget over keys; (ii) a horizon accountant for repeated lookups, yielding a profiling label that is OINL-ready (Synthesis 0; Anonymity C).[10, 17]

Claim fields (non-negotiable). Every bound in this paper is parameterized by `exposure_nf_id` (ENF-ID) naming X and by `tw_id` (TW-ID) naming the repetition window and active/passive template. For stateful deployments, the accountant may additionally depend on a digest-bound `state_decl_id`; when that happens, `state_decl_id` is a third claim field and drift semantics are handled by the plan/state equivalence note.[14] If either id changes, the resulting profiling budget is a different claim even if the numeric (η, δ) values match.[11, 12]

Notation. We follow the series notation sheet (Synthesis 8).[32] K is the hidden key (or key-class), T is the raw transcript, and $X = \pi(T)$ is the declared projection seen by the adversary tier.

Artifact policy. Source-only; supporting logs, scripts, and certificates are available on request as a single bundle committed by ABOM/OINL (Anonymity C) under the series artifact policy.[17, 1]

How to read this in the series. If you want (i) the endpoint meaning of the bounds, read Anonymity A; (ii) interface vocabulary, read Synthesis 3; (iii) horizon semantics and accounting, read Synthesis 4 and Synthesis 19; and (iv) the publication object, read Anonymity C.[2, 11, 12, 13, 15, 17]

1 Profiling as repeated testing

A structured overlay lookup (e.g., Kademlia[20]) induces an observable transcript: which peers are contacted, message sizes, timing buckets, abort/retry decisions, congestion witnesses, and any public control-plane beacons. Even if single lookups are “approximately anonymous,” *repeated* lookups can be identifying. Three canonical examples:

1. **Content systems:** IPFS-style content routing demonstrates that DHT-facing metadata enables large-scale passive monitoring and interacts with censorship/centralization pressures; for concrete measurements and privacy threats (including passive monitoring methodology), see Balduf et al.[28] and the broader comparators in the related-work map (Synthesis 7).[27, 33] For deployed control-plane surfaces (delegated-routing endpoints, AutoConf, OHTTP upgrades), see Synthesis 30.[34]
2. **Anonymous lookup lineage:** attacks and information leaks in DHT-based anonymity systems motivate specialized anonymous lookups (split queries, cover, delegation) and query-privacy primitives, and more recent work explores practical obfuscation layers for Kademlia-style lookups.[23, 21, 22, 24, 29]
3. **Network-scale measurement:** real-world DHTs have been profiled at million-node scale, making transcript aggregation operationally feasible.[25]

For additional comparators and concrete measurement studies, see the related-work map (Synthesis 7).[33]

We model the profiling adversary as seeing n (possibly dependent) samples $X_1, \dots, X_n$ of a chosen projection X of the full transcript, and attempting to infer a hidden index $K \in \{1, \dots, |\mathcal{K}|\}$ (a key, destination class, or content ID bucket).

We emphasize that these concerns persist in newer “anonymous DHT” proposals for other settings (e.g., blockchain overlays) when deployments expose repeatable transcript features; the equalization contract is meant to be portable across such designs.[26]

2 The equalization contract

Definition 1 (Equalization target). *Fix a measurable transcript projection X (examples below). For each hidden key $k \in \mathcal{K}$, let P_k denote the law of X for a single lookup to k under the deployment’s policy.*

Definition 2 (Pairwise equalization). *A deployment satisfies pairwise (η, δ) equalization for X if for all $k, k' \in \mathcal{K}$ and all measurable E ,*

$$P_k(E) \leq 2^\eta P_{k'}(E) + \delta.$$

This is a max-divergence style statement (an (η, δ) stability guarantee) restricted to the X -projection.

Remark 1 (Equalization is only as good as the declared projection). *The choice of X is not purely mathematical: deployed control-plane surfaces (router sets, endpoint autoconfiguration, cache layers, retrieval-mode switches) can change what is observable even when the “mechanism” is unchanged. In this series, the projection choice is therefore coupled to (i) an exposure normal form id (Synthesis 3) and (ii) a digest-bound state declaration that pins the relevant surfaces under change control (Synthesis 18 and Synthesis 30). [11, 14, 34]*

Lemma 1 (Pairwise equalization $\Rightarrow$ one-vs-mixture anonymity for X). *Fix any prior π on $\mathcal{K}$ and let $P := \sum_{j \in \mathcal{K}} \pi(j) P_j$ be the public marginal of X . If pairwise (η, δ) equalization holds, then for every $k \in \mathcal{K}$ and every measurable event E ,*

$$P_k(E) \leq 2^\eta P(E) + \delta.$$

In particular, when $\delta = 0$ we have the pointwise likelihood-ratio cap $p_k(x)/p(x) \leq 2^\eta$ for all x with $p(x) > 0$. Equivalently, the realized odds inflation of the X -projection (Anonymity A) is at most η .

Proof. Multiply the pairwise inequality by $\pi(k')$ and sum over $k' \in \mathcal{K}$. The left-hand side stays $P_k(E)$; the first right-hand term becomes $2^\eta \sum_{k'} \pi(k') P_{k'}(E) = 2^\eta P(E)$; and the additive term sums to δ . For $\delta = 0$, apply the same argument pointwise to the singleton event $\{X = x\}$. $\square$

Remark 2 (Why max-divergence?). *Total variation constraints are often natural (especially when X is discrete and bounded), but TV alone does not control pointwise likelihood ratios. Max-divergence and odds-inflation are “endpoint friendly”: they translate directly into posterior ratio bounds and into the identification sizing rules used throughout this series. [2, 3] When a mechanism is designed/verified in TV (e.g., via transport dual certificates in the published Knapsack/Transport note [19]), we treat the TV $\rightarrow$ odds conversion as an imported module (Synthesis 5).*

2.1 What can be equalized? (examples)

The contract is intentionally modular: different papers “own” different X projections (see the projection normal forms in Synthesis 3). [11] Examples:

- **Committee-contact footprints (MUCC).** X is the multiset of committee IDs contacted in each epoch. (AnonDHT State 2.)
- **Termination-time buckets.** X is a deadline bucket / stop time normal form for a lookup. (Published 2026-01-26 wiki post.)

- **Prefix-fetch witnesses.** X is a prefix-length trace (or a coarsened prefix-set witness) induced by *prefix fetching* (widened lookups). Prefix-level equalization is constrained by shared-kernel termination-time lower bounds (Prefix Capacities, 2026-01-25) and by stop-time normal forms (2026-01-26); treat feasibility and “what can actually be made key-independent as imported math, and expose only the resulting budgeted projection.[38, 18]
- **Congestion witnesses.** X is the RTT/queueing witness used by Congestion-EQ. (Congestion series.)
- **Retry traces.** X is the schedule-only retry trace X_{retry} exported by the schedule-only wrapper (Operational A).[4]
- **Receipts / certificates.** X is the published claim object itself (receipt redaction bounds). (Synthesis 1; Release A; W-Congestion-EQ.)

The key point: you do *not* need to equalize the entire transcript to obtain a meaningful profiling budget. You only need to equalize what the adversary can reliably observe and aggregate.

3 From equalization to profiling budgets

Lemma 1 turns pairwise (η, δ) equalization for a projection X into a one-vs-mixture (η, δ) anonymity bound for X . To obtain a profiling statement over repeated lookups, we then compose this one-step bound over the repetition horizon as a *horizon claim* (unit+reset+accountant+replay hook) per the receipt-accounting rulebook.[13, 12] Rather than re-derive composition here, we import the sequential (ε, δ) “checkpoint” calculus from the odds-inflation endpoint paper.[2]

Horizon- n profiling label (imported composition). This paper does not re-derive horizon composition. Apply the sequential (ε, δ) checkpoint composition theorem from *Odds Inflation* (Anonymity A) to the per-lookup one-vs-mixture inequality obtained in Lemma 1.[2] Concretely, suppose the deployment executes n lookups inside the declared threat window and publishes per-lookup equalization budgets (η_ℓ, δ_ℓ) for the same projection family X . If, almost surely over the declared public history/state in scope, each step satisfies the conditional inequality

$$P_k(X_\ell \in E \mid h) \leq 2^{\eta_\ell} P(X_\ell \in E \mid h) + \delta_\ell \quad \text{for all keys } k \in \mathcal{K} \text{ and events } E,$$

then the joint transcript $X_{1:n}$ satisfies a one-vs-mixture anonymity bound

$$(\eta_\star, \delta_\star) = \left(\sum_{\ell=1}^n \eta_\ell, \sum_{\ell=1}^n \delta_\ell 2^{\sum_{j<\ell} \eta_j} \right).$$

In the homogeneous case $(\eta_\ell, \delta_\ell) = (\eta, \delta)$, this simplifies to

$$(\eta_\star, \delta_\star) = (n\eta, \delta \sum_{j=0}^{n-1} 2^{j\eta}).$$

When retries/caching/adaptive routing break conditional independence, the same conditional-checkpoint formulation still applies; the receipt-facing condition is carried by the threat-window and (if needed) a digest-bound state declaration.[12, 5, 14]

3.1 Conditioning on published witnesses (state-conditioned equalization)

Some deployments publish a coarse state witness W (e.g., a public staleness class or rate-limit class) and certify equalization of X conditional on W . Formally, require that for every witness value w , the conditional laws $P_k^{X|W=w}$ satisfy pairwise (η, δ) equalization. If W is public and key-independent (as declared by a digest-bound `state_decl_id`), then controlling the worst-case conditional leakage of X given W controls the joint leakage of (W, X) ; see the conditioning lemma in the plan/state equivalence note.[14] In receipts, the witness can be embedded into the ENF conditioning fields or exported as an explicit Tier 0 line item; conditional certificates can be carried as Regime II records.[15]

Identification cap (OINL-facing). The endpoint bridge converts $(\eta_\star, \delta_\star)$ into an explicit identification cap.[3] Let $K \sim \pi$ and let $\widehat{K}(X_{1:n})$ be any estimator. Then

$$\Pr[\widehat{K} = K] \leq 2^{\eta_\star} \pi_{\max} + \delta_\star, \quad \pi_{\max} := \max_{k \in \mathcal{K}} \pi(k),$$

and under a uniform prior on $|\mathcal{K}|$ keys, $\Pr[\widehat{K} = K] \leq 2^{\eta_\star} / |\mathcal{K}| + \delta_\star$.

OINL-ready profiling label. For each declared projection X , publish (i) the per-lookup equalization budget(s) (η_ℓ, δ_ℓ) , (ii) the threat-window horizon n (or a time window plus a lookup-rate assumption), and (iii) the composed pair $(\eta_\star, \delta_\star)$ together with the identification cap above. ABOM/OINL specifies how to package this claim and commit to its supporting artifacts by digest.[17]

Contract object (minimal, non-redundant). Publish the equalization claim as a small contract object (Synthesis 0):[10]

- **Projection id + schema:** a stable name for X (ENF-ID backed) and a schema digest for the extractor $f(T)$. [11]
- **Secret scope:** what K ranges over (keys, key-classes, destinations) and what eligibility/conditioning is treated as public.
- **Budget + horizon:** (η_ℓ, δ_ℓ) and the TW declaration used to determine n ; if state enters, include `state_decl_id`. [12, 14]
- **Evidence pointer:** proof objects and/or audit receipts + the ABOM digest that bundles omitted artifacts. [17]

Multiple projections. Deployments often publish several equalization claims (retry trace, committee-contact footprint, congestion witness, stop-time bucket). Treat each as a separate witness tier and compose using the series contract-object rulebook (Synthesis 0) rather than re-deriving a global guarantee here.[10]

4 Certifying equalization from logs (brief)

A production workflow needs a way to certify the equalization claim. This series provides two complementary levers:

1. **Mechanism-first certificates.** When X is generated by a compiler with a known proof object (e.g., TV transport duals, or proof-carrying padding schedules), publish the certificate (BossFight C; the 2026-01-29 wiki post).
2. **Data-first audits.** When only logs are available, treat equalization as a two-sample problem (compare empirical $\hat{P}_k$ across k), but correct for *many-testing* across many keys and policies. Synthesis 25 provides menu-level simultaneous guarantees for discrete projections audited from logs, and Synthesis 26 standardizes how a deployment spends a single δ_{stat} over repeated audits or expanding menus.[6, 9]

Log-audit certificates (pointer). When only logs are available, equalization can be audited as a finite-sample distributional claim on a chosen interface projection X . Rather than duplicating statistics across papers, this archive centralizes “menu-level” log-audit bounds (including many-testing correction across many keys/policies, and receipt-side fields for publishing δ_{stat} and sample sizes) in Synthesis 25.[6]

In brief, an auditor chooses (i) a discrete projection X (bucketed time / contact-set ID / retry-pattern hash / prefix-length bin, . . .), (ii) a confidence level δ_{stat} , and (iii) a scope (which keys/policies are in the menu). If the projection depends on a data-selected bucket/coarsening map, apply the selection discipline note (Synthesis 29): treat the map family as witness-indexed or separate pilot selection from frozen certification under change control.[39] From empirical frequencies $\hat{P}_k$ per class k , Synthesis 25 provides a conservative bound of the form $\text{TV}(P_k, P_{k'}) \leq \text{TV}(\hat{P}_k, \hat{P}_{k'}) + r_k + r_{k'}$ holding simultaneously over the declared menu. Publishing the resulting bound as a **profiling_evidence** object is then enough to support the equalization contract, while keeping the paper itself referee-grade and short. For a fully wired receipt+bundle instantiation (including the prefix-fetch variants), see Synthesis 17.[31] For a concrete coarsening instantiation (large-alphabet projection made audit-feasible by a declared partition map), see the coarsening variant receipt in Synthesis 17 (and the design rule in Synthesis 28).[7]

5 Positioning (one paragraph)

This note does not propose a new anonymous-lookup protocol. It isolates a reusable *publication contract*—equalization on a declared transcript projection X —and a reusable *budget consequence*—profiling bounds via odds inflation and identification caps. Protocol mechanisms and measurement studies live in the related-work map (Synthesis 7), which this paper treats as the canonical pointer list.[33]

Active routing attacks (Sybil/eclipse) can also *amplify* profiling by shaping contact sets and termination behavior; in such regimes the equalization contract may fail even if the benign-network contract is certified. Deployments should therefore treat routing-security posture as part of the declared state surface, and (when relevant) publish a separate “under-attack” budget or explicitly scope it out.[41, 42]

6 Takeaways

The “anonymity” question for DHT lookups is often too blunt. Profiling is driven by concrete, repeatable skews in a small number of observable features. An equalization contract turns those skews into budgets, and the series’ odds–inflation bridge turns budgets into the quantities operators and downstream users can reason about (posterior inflation and identification scale).

7 Where more work is warranted (short list)

- **Equalization under dependence.** The clean horizon scaling in the horizon composition bound above assumes conditional independence. Caches, adaptive routing, and key-dependent retries create stateful dependence; bridging “equalization contracts” to the state-series odometers in a way that remains OINL-friendly is a high-leverage next step. For random-walk delegation, Anonymity D gives a compact spectral-witness pathway to Regime II per-step certificates.[5, 15, 16]
- **Which projection to equalize (cost vs. value).** Tier 0/1 equalization can be cheap but may leave Tier 2 failure-order traces exploitable. A systematic “value of coarsening” study—how much profiling power survives when moving from Tier 2→Tier 1→Tier 0 for real overlays—would inform what ENF-IDs are worth certifying.[11]
- **Drift and manipulation of the background Q .** When Q is measured (not specified), the measurement process itself is an attack surface. A lightweight, notarized drift test that feeds directly into TW declarations (what window it applies to) and ABOM change control (what recertification is triggered) is still missing as a single, reusable module.[12, 13, 17]

References

- [1] Anonymous. Artifact and Disclosure Policy for the One-True-Archive (Source-Only Public Release). Series synthesis note (Synthesis 6), 2026. (This archive.)
- [2] Anonymous. Odds Inflation, Privacy Loss, and Endpoint Anonymity Bounds. Anonymity series Paper A, 2026. (This archive.)
- [3] Anonymous. Endpoint Metrics Bridge: Odds Inflation, PML/MaxL, and Identification Sizing. Series synthesis note (Synthesis 5), 2026. (This archive.)
- [4] Anonymous. Schedule-Only Retries in Anonymous DHTs: SRP and Retry-Trace Equalization. Operational series Paper A, 2026. (This archive.)
- [5] Anonymous. State-Dependent Anonymity in Anonymous DHTs. AnonDHT state series Paper 1, 2026. (This archive.)
- [6] Anonymous. Discrete Equalization Audits from Logs: Menu-Level Finite-Sample TV Certificates. Synthesis 25, The-One-True-Archive (2026).
- [7] Anonymous. Coarsening and Large-Alphabet Projections for Log Audits: Safe Partitioning Rules that Keep Certificates Sample-Feasible. Synthesis 28, The-One-True-Archive (2026).
- [8] Anonymous. Bucket Design for Continuous Timing Interfaces: Making “Continuous” Telemetry Auditably Discrete. Synthesis 27, The-One-True-Archive (2026).
- [9] Anonymous. Anytime Statistical Spending for Menu Audits: Change-Control-Safe Confidence Budgets for Expanding Log Certificates. Synthesis 26, The-One-True-Archive (2026).
- [10] Anonymous. Contract Objects for Anonymous-DHT Privacy Budgets and Claims. Series synthesis note (Synthesis 0), 2026. (This archive.)
- [11] Anonymous. Trace Interfaces for Anonymous DHTs: Observation Tiers, Committee-Contact Traces, and Exposure Normal Forms. Series synthesis note (Synthesis 3), 2026. (This archive.)
- [12] Anonymous. Threat Windows for Anonymous DHTs: What a Budget Covers and For How Long. Series synthesis note (Synthesis 4), 2026. (This archive.)

- [13] Anonymous. Receipt Accounting and Horizon Composition for Anonymous-DHT Budgets. Series synthesis note (Synthesis 19), 2026. (This archive.)
- [14] Anonymous. Digest-Bound Replay Plans and State Declarations: Equivalence, Minimality, and Change Control. Series synthesis note (Synthesis 18), 2026. (This archive.)
- [15] Anonymous. Conditional Per-Step Budget Certificates: A Receipt-Grade Interface for Adaptive Horizon Claims. Series synthesis note (Synthesis 21), 2026. (This archive.)
- [16] Anonymous. Spectral Delegation for Anonymous DHT Lookups: From mixing certificates to receipt-grade per-step budgets. Anonymity series Paper D, The-One-True-Archive (2026).
- [17] Anonymous. ABOM/OINL: Audit BOM and Outcome Identification Noise Label for Anonymous DHT Deployments. Anonymity series Paper C, 2026. (This archive.)
- [18] Anonymous. Stop-Time Padding Addendum (unique-only): Max-Mixture Separations and Tail-Sign Deadline Normal Forms. Series note (published), 2026. Wiki: `[[2026.01.26 - Mathematics: Stop-Time Padding Addendum, Max-Mixture Separations and Tail-Sign Deadline Normal Forms]]`.
- [19] Anonymous. Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy. Series note (published), 2026. Wiki: `[[2026.01.29 - Mathematics: Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy]]`.
- [20] P. Maymounkov and D. Mazieres. Kademlia: A peer-to-peer information system based on the XOR metric. In *IPTPS*, 2002.
- [21] P. Mittal and N. Borisov. Information leaks in structured peer-to-peer anonymous communication systems. *ACM Transactions on Information and System Security*, 15(1), 2012.
- [22] Q. Wang and N. Borisov. Octopus: A secure and anonymous DHT lookup. In *ICDCS*, 2012.
- [23] Q. Wang, P. Mittal, and N. Borisov. In search of an anonymous and secure lookup: attacks on structured peer-to-peer anonymous communication systems. In *ACM CCS*, 2010. (Unpaywalled PDF: freehaven anonbib.)
- [24] M. Backes, I. Goldberg, A. Kate, and T. Toft. Adding query privacy to robust DHTs. In *ASIACCS*, 2012. DOI: 10.1145/2414456.2414473. (Also available as arXiv:1107.1072.)
- [25] J. Falkner, M. Piatek, J. P. John, A. Krishnamurthy, and T. Anderson. Profiling a million user DHT. In *ACM IMC*, 2007.
- [26] L. Tseng, L. Ambarapu, and M. Aloqaily. NC-DHT: a robust and anonymous DHT for blockchain systems. In *2024 6th International Conference on Blockchain Computing and Applications (BCCA)*, IEEE, 2024, pp. 394–399. DOI: 10.1109/BCCA62388.2024.10844445.
- [27] D. Trautwein, A. Raman, G. Tyson, I. Castro, W. Scott, M. Schubotz, B. Gipp, and Y. Psaras. Design and Evaluation of IPFS: A Storage Layer for the Decentralized Web. In *ACM SIGCOMM*, 2022, pp. 739–752. DOI: 10.1145/3544216.3544232. Preprint: arXiv:2208.05877. <https://arxiv.org/abs/2208.05877>.
- [28] L. Balduf, S. A. Henningsen, M. Florian, S. Rust, and B. Scheuermann. Monitoring Data Requests in Decentralized Data Storage Systems: A Case Study of IPFS. In *IEEE International Conference on Distributed Computing Systems (ICDCS)*, 2022, pp. 658–668. DOI: 10.1109/ICDCS54860.2022.00069. Preprint: arXiv:2104.09202. <https://arxiv.org/abs/2104.09202>.
- [29] E. Daniel, G. Michel, and F. Tschorsch. Improving Kademlia Lookup Privacy through Query Obfuscation. In *ACM/SIGAPP Symposium on Applied Computing (SAC)*, 2025, pp. 1345–1352. DOI: 10.1145/3672608.3707703. Record: <https://fis.tu-dresden.de/portal/en/publications/improving-kademlia-lookup-privacy-through-query-obfuscation%2834f33c16-7610-4c43-9ba1-028fd028f033%29.html>.

- [30] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Series synthesis note (Synthesis 12), 2026. (This archive.)
- [31] Anonymous. Worked Example: Instantiating a Tiered Reader-Privacy Receipt. Series synthesis note (Synthesis 17), 2026. (This archive.)
- [32] Anonymous. Notation and Minimal Model for the One-True-Archive. Series synthesis note (Synthesis 8), 2026. (This archive.)
- [33] Anonymous. Related Work Map for Anonymous DHT Lookups. Series synthesis note (Synthesis 7), 2026. (This archive.)
- [34] Anonymous. Deployed Observation Surfaces for Anonymous DHTs: Control-plane knobs in IPFS/libp2p delegated routing and OHTTP evolution. Series synthesis note (Synthesis 30), 2026. (This archive.)
- [35] Anonymous. AnonDHT in the Boss Fight Regime: Compiling a MaxL Budget into Lookup Knobs. Boss Fight series Paper B, 2026. (This archive.)
- [36] Anonymous. Notarized Anonymity Certificates: Release-Bound Manifests and Anti-Grinding Discipline. Evaluation series Paper 1, 2026. (This archive.)
- [37] Anonymous. Advantage Contracts and Stealth Audits for Anonymous-DHT Privacy Claims. Evaluation series Paper 2, 2026. (This archive.)
- [38] Anonymous. Prefix Capacities and Separation Cuts for Shared-Kernel Termination-Time Hiding. Published mathematics note, 2026-01-25. (This archive: [published/2026-01-25_prefix_capacities.](#))
- [39] Anonymous. Selection Discipline for Data-Dependent Bucket Maps. Series synthesis note (Synthesis 29), 2026. (This archive.)
- [40] M. Mazmudar, S. Veitch, and R. Akhavan Mahdavi. Peer2PIR: Private Queries for IPFS. In *IEEE Symposium on Security and Privacy (S&P)*, 2025. DOI: 10.1109/SP61157.2025.00231. Preprint: arXiv:2405.17307 (2024). <https://arxiv.org/abs/2405.17307>.
- [41] V. H. M. Netto, T. Cholez, and C.-L. Ignat. Active Sybil Attack and Efficient Defense Strategy in IPFS DHT. arXiv:2505.01139, 2025. <https://arxiv.org/abs/2505.01139> (accessed 2026-03-04).
- [42] Thibault Cholez and Claudia-Lavinia Ignat. Sybil Attack Strikes Again: Denying Content Access in IPFS with a Single Computer. In *ARES 2024*. DBLP record: <https://dblp1.uni-trier.de/rec/conf/IEEEares/CholezI24.html> (accessed 2026-03-04).